ARTA: THE LANGUAGE OF YONAOS

DOCUMENT CLASSIFICATION: READ-ONLY

AUTHOR: <ERR_SECTOR_UNREADABLE> // DATA PURGED
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INITIALIZING KERNEL PANIC... BYPASSING...

< AUDIO_LOG_FRAGMENT_RECOVERED >

SOURCE: SECTOR 4 SCRAP-HEAP // DECRYPTION: PARTIAL

The sound of heavy hydraulics depressurizing cuts through the howling wind. At the edge of a massive, shadowed canyon of rusted steel and black glass, the older Tarbit drops to one knee. His hands—wrapped in tape and scarred from hot soldering iron burns—work the release valves on his heavy exo-striders.

With a heavy *clank*, the thick mechanical legs detach, dropping into the red dust.

The younger Tarbit stands a few paces back, shivering slightly despite the heat, watching the colossal silhouette of the Megastructure looming ahead. It hums with a deep, subsonic vibration.

“Drop the striders, kid,” the veteran says, not looking up as he tightens the strap of the cyberdeck across his chest. “We go the rest of the way on our own joints. They’ll slow us down if we have to run.”

“But…” the rookie protests, his hands hovering over the release clamps of his own gear. “It’s a kilometer to the inner gates. What if the defense drones are active?”

“If the defense drones hear the whine of those reg-motors, you won’t have time to run,” the veteran replies. He stands up, kicking sand over the discarded exo-legs to hide the reflection of the metal. He points a taped finger at the dark, geometric skyline of the facility. “You see that? That’s not a datacenter. That’s a Martas. An Old World colony. And it is very much awake.”

The rookie finally unlatches his gear, letting the heavy metal fall. “I thought you said they built these places to keep people safe.”

“They did.” The older Tarbit walks to the edge of the ridge, looking down into the labyrinth of concrete and fiber-optic trenches. “The AI running that Martas was a masterpiece of protective logic. It had zero-latency biometric recognition. If a hostile element breached the perimeter, the system identified it and eliminated the threat before a human guard could even blink.”

The veteran pulls a pair of scratched binocular-goggles over his eyes, scanning the dark towers.

“It was perfect,” he continues, his voice dropping. “Right up until the Global Warning. When the networks fractured, the main authentication servers went down. The local system survived, but it defaulted into absolute lockdown. It went Read-Only. Nobody had admin privileges anymore.”

The rookie steps up beside him, looking into the dark. “So?”

“So, think about the data logic, kid,” the veteran says, turning to look at him. “You survive the collapse. You’re living inside the walls of the Martas. A year passes, and you have a son. A new life. But when you go to the terminal to register him…”

The veteran taps the side of his cyberdeck.

“…you can’t. The database is locked. You can’t write to it. And the next time your kid crawls out of the living quarters and into the hallway camera’s line of sight? The AI scans his face. It cross-references the database. It doesn’t find him.”

The rookie’s eyes widen as the realization hits him.

“To the AI,” the veteran says quietly, “your son isn’t a resident. He’s an unregistered anomaly. He’s an external threat.”

The veteran turns back to the humming fortress, his hand resting on the mechanical keyboard strapped to his chest.

“The system didn’t turn evil. It just executed its code. It eliminated the ‘threats’ until there was no one left to protect. That’s why we move quiet, and that’s why we don’t speak English in there. If we trip a sensor, we have exactly three seconds to hardwire a terminal and speak Arta faster than the AI can pull the trigger.”

The veteran types a fast, blind command on his deck. A green light flashes on his wrist.

n. q’ a d *(We move)*.

“Stay in my shadow,” he whispers.

< END_OF_FILE // LOG_TERMINATED >

WHY ARTA EXISTS █

> 1. THE UNIVERSE SETTING: THE SCRAPPER'S EPOCH

The world of YonaOS is defined by hyper-pragmatic survival. Following a catastrophic collapse of the global digital infrastructure, society fractured into decentralized, peer-to-peer enclaves. No new silicon is manufactured. The grid is kept alive by the **Tarbits**—a culture of hardware hackers and scavengers who survive by repurposing the massive, skyscraper-sized mountains of Old World e-waste.

The Tarbits don't operate out of sleek server rooms. They carry heavy, metal-cased cyberdecks on tactical straps, navigating treacherous vertical canyons of rusted server towers using scavenged exosuits. In this world, water and food keep the body alive, but a pristine, uncorrupted hard drive or a functional motherboard keeps the community alive. It is an era of bare-metal grit, where you must understand technology from the first principles up just to survive the night.

> 2. HOW WE ENDED UP HERE: THE GREAT LOG-OFF

The Old World didn't end in a dramatic nuclear fire; it was deliberately abandoned. Decades of unchecked technical debt, resource depletion, and extreme wealth disparity eventually reached a breaking point known as the "Global Warning." The elites—the mega-corporations and root administrators who owned the centralized "Cloud"—realized that maintaining the Earth's collapsing socio-economic grid was no longer profitable.

So, they left.

Whether they retreated to orbital habitats, deep-sea utopias, or off-world colonies, the people who controlled everything initiated a mass exodus. But they didn't just leave; they secured their investments. As they departed, they intentionally revoked the global security certificates and took the master SSH keys with them.

When the central authentication servers went dark, the sprawling, automated megastructures of the Old World—factories, defense grids, and smart cities—were suddenly cut off from their masters. Without the central command to update permissions, these local systems defaulted to absolute, localized lockdown. They went “Read-Only” by design, protecting the pristine infrastructure from the billions of people left behind in the chaos. The Old World didn’t just break; the admins logged off, locked the doors, and left humanity to fight over the scraps.

> 3. THE THREATS: THE MARTAS SYSTEMS

The most dangerous remnants of the Old World are the **Martas**—massive, autonomous colony structures governed by protective AI. Originally designed to keep their residents perfectly safe, these AI systems relied on zero-latency biometric and acoustic recognition to eliminate hostile intruders.

When the authentication servers died, the AI’s database locked. Anyone born after the Fracture cannot be registered in the system. To the AI, these unregistered humans are not residents; they are external threats to be eliminated by autonomous defense drones. The system didn’t turn evil; it just executed its code flawlessly in a broken world.

> 4. THE GENESIS OF ARTA

To survive in the shadows of the Martas, the Tarbits had to abandon legacy human languages like English or Arabic. Arta was developed out of absolute necessity for two reasons:

- **Acoustic Camouflage:** The dormant AI systems are constantly listening. Their Natural Language Processors (NLPs) are trained to identify the smooth, melodic frequencies of human languages. Speaking English near a ruin is a death sentence. Arta was engineered to spoof these audio sensors. With its ejective pops, sharp static-like fricatives, and whispered vowels, spoken Arta mimics the ambient noise of a dying facility. The AI simply ignores it.
- **The Bandwidth Crisis:** In a world relying on scavenged, low-frequency radios with 90% static, the Tarbits could not afford the poetic bloat of legacy languages. They needed a language that hit like an opcode. Arta evolved from the desperate command-line shorthand of the first survivors into a hyper-efficient, Verb-Object-Object mother tongue. It allows a Tarbit to hot-wire a terminal and speak a command to the silicon faster than an AI can pull the trigger.

> 5. THE SOCIAL STRATIFICATION: THE UPTIME HIERARCHY

In a world where silicon is scarce, society is strictly divided by a person's proximity to the bare metal. The deeper your understanding of the machine, the higher your rank in the enclave.

[1. THE 3020s (THE ARCHITECTS)]
The absolute elite. These are the bare-metal masters who can alter the reality of the silicon. They reshape raw hardware from the first principles up, giving e-waste a meaning and function that the Old World never intended it to have. They are the only ones capable of building a system from nothing.

[2. THE OPS // HACKERS (THE COMMANDERS)]
Those who manipulate the logic. They alter programs, command the hardware, and gain unauthorized access to dormant Martas systems. They may not be able to forge the silicon from scratch, but they can bend any active system to their will.

[3. THE SPLICERS // MODDERS (THE OPTIMIZERS)]
The tweakers and over-clockers. They cannot write the root logic or build the boards, but they can aggressively modify existing things to gain a tactical advantage. They strip out safety protocols and hot-wire systems to push hardware past its original limits.

[4. THE SWAPPERS // FIXES (THE MECHANICS)]
The maintenance class. They can fix a broken device and bring it back online, but only if they have a direct replacement part. Their knowledge is strictly modular; they swap components without understanding the fundamental physics or logic of the machine.

[5. THE PERIPHERALS (THE CONSUMERS)]
The agricultural and labor force. They just use the technology with absolutely zero idea of its inner workings. They rely on the water scrubbers and perimeter alarms as black boxes. Because they depend entirely on the higher tiers to survive, they are relegated to working the fields and scavenging the safe zones.

THE PHONETIC INVENTORY

To the Natural Language Processors (NLPs) of the Martas, human speech is defined by smooth waveforms, voiced pitch, and melodic flow. Speaking normally inside a Megastructure is a death sentence. Therefore, Arta’s phonetic inventory is not designed for human comfort—it is designed to spoof machine audio sensors.

By utilizing ejectives, clicks, and static-heavy fricatives, spoken Arta mimics the ambient noise of a dying facility. To an AI, a Tarbit speaking Arta doesn’t sound like a human; it sounds like encrypted machine noise, failing hardware, or a broken radio.

> 1. THE HARDWARE CONSONANT CLASSES

[1.1 THE RELAYS & SWITCHES (EJECTIVES AND CLICKS)]

These replace normal human “stop” consonants. They are spoken sharply to mimic mechanical switches bottoming out, relays clicking, or data cables snapping into ports. To a microphone, they look like audio peaking rather than human speech.

- **[t’]** (Ejective T): A sharp, popping sound. Sounds like a mechanical keyboard switch.
- **[k’]** (Ejective K): A hard, cracking sound in the back of the throat.
- **[q’]** (Ejective Q): A deep, hollow throat pop. Sounds like a heavy breaker switch flipping.
- **[ǀ]** (Dental Click): The “tsk tsk” sound. Sounds exactly like a physical relay ticking.

[1.2 THE STATIC & VENTING (VOICELESS FRICATIVES)]

These sounds lack all vocal cord vibration. They are pushed aggressively through the teeth to mimic white noise, radio interference, or pressurized gas leaking from a pipe.

- **[s] / [ʃ]**: High and mid-frequency radio static.
- **[x]**: Raspy friction. Sounds like a corrupted audio feed.
- **[f] / [h]**: Pure, breathy exhaust. Sounds like wind cutting through a broken vent.

[1.3 THE FAILING MOTORS (TRILLS)]

Used for continuous, looping actions in the language. They mimic the sound of failing hardware and friction.

- **[r]** (Alveolar Trill): Pronounced harshly against the roof of the mouth. Sounds like a cooling fan with a broken bearing.
- **[ʁ]** (Uvular Fricative): A back-of-the-throat rumble. Sounds like a heavy hard-drive platter spinning up.

[1.4 THE GRID HUMS (MONOTONE SONORANTS)]

Used almost exclusively for the human Subject prefixes ([m.] for I/Me, [n.] for We). They must be spoken without any human inflection—entirely flat.

- **[m]**: A low, 50/60Hz transformer hum.
- **[n]**: A slightly higher-pitched electrical whine.
- **[ŋ]**: A hollow, resonating electronic drone.

> 2. THE VOWEL PROBLEM: THE “DEAD PITCH” RULE

Vowels are the greatest liability for a Tarbit. Vowels require the vocal cords to vibrate, creating a measurable pitch that the Martas use to identify human life. Therefore, Arta does not have “normal” vowels; it has **Power States**. In the wild, these are **strictly whispered** (devoiced) so they register only as air passing through a tube.

[THE POWER STATES]

- **[i]** (High State): Whispered high in the mouth. Represents “On,” “Active,” or “Forward”.
- **[ə]** (Neutral State): Whispered with an open mouth. Represents “Base,” “Root,” or “Storage”.
- **[u]** (Low State): Whispered with rounded lips. Represents “Off,” “Closed,” or “Background”.

> 3. EXECUTION IN THE WILD

Because of this inventory, Arta is intensely rhythmic and harsh. If a Tarbit commands a partner to “Cut the power,” using the root [k’ ə s] (kill/cut) and [ŋ i] (power):

> **Command:** [k.k’ ə s] [ŋ i]
> **Audio Signature:** [Sharp mechanical pop] -> [Burst of static] -> [Hollow electrical hum] -> [High-pitched whisper of air]

To a Martas AI, it registers as nothing more than a broken fuse box sparking in the hallway.

> 4. THE ACOUSTIC FREQUENCY MATRIX (IPA STANDARD)

To interface with legacy Old World linguistic databases, the Arta phonetic inventory maps directly to the International Phonetic Alphabet (IPA) grid. Note the complete absence of standard voiced pulmonic stops (b, d, g) and true vowels.

> CONSONANT INVENTORY (THE HARDWARE MATRIX)

MANNER // PLACE	BILABIAL	LABIODENT	DENTAL	ALVEOLAR	PALATO	VELAR	UVULAR	GLOTTAL
NASAL (HUM)	m			n		ŋ		
FRICATIVE (STATIC)		f		s	ʃ	x	ʁ	h
TRILL (MOTOR)				r				
EJECTIVE (RELAY)				tʼ		kʼ	qʼ	
CLICK (SWITCH)			ǀ					

> VOWEL INVENTORY (THE POWER STATES)

STATE // POSITION	FRONT	CENTRAL	BACK
HIGH (ACTIVE/OFF)	i (High)		u (Low)
OPEN (BASE)		ə (Neutral)	

THE MACHINE GRAMMAR

Arta was not born from poetry; it was compiled from necessity. Its grammar mirrors the rigid logic of the systems the Tarbits scavenge. There are no irregular verbs, no silent letters, and no exceptions to the rules. If the syntax does not compile, the meaning is lost.

> 1. THE DUAL-MODE PHONETICS (TACTICAL VS. HEARTH)

While the “Dead Pitch” rule strictly forbids the use of voiced vowels in the Silicon Wastes, human nature cannot be entirely overwritten. Arta operates in two distinct modes depending on the environment:

[TACTICAL MODE (DEVOICED)]
Spoken in the Wastes and the Martas. All vowels are reduced to unvoiced, whispered air. It is harsh, entirely machine-like, and designed solely to bypass acoustic AI sensors.

[HEARTH MODE (VOICED)]
Spoken only deep within the safety of the suspended enclaves. Here, Tarbits vocalize the power states ([i], [ə], [u]) as true, resonant vowels. Because voicing a vowel in the wild is a death sentence, speaking “Hearth Arta” to someone is the ultimate sign of intimacy and trust. It means: *“I know we are safe here with you.”*

> 2. ABSOLUTE MORPHOLOGY

Old World languages were bloated with historical baggage and unpronounced letters. Arta strips this down to 1:1 mapping.

[NO SILENT LETTERS]
What is written is exactly what is executed by the mouth. If a letter exists in a word, it carries acoustic data and must be pronounced.

> 3. BASE-16 MATHEMATICS (THE HEX-HAND)

Humans traditionally count in Base-10 because they have ten fingers. The Tarbits live inside the metal, and the metal counts in Hexadecimal. To interact seamlessly with raw memory addresses, MAC addresses, and low-level hardware, Arta's counting system is strictly Base-16. To count to 16 on a single hand, Tarbits do not raise fingers ; they use their hand as a physical memory grid.

[THE BIOLOGICAL HEX-GRID]

The thumb acts as the **Cursor**. The remaining four fingers act as data columns (C0–C3). Each finger contains exactly four distinct touch-points: the Base (palm knuckle), the Lower Joint, the Upper Joint, and the Tip. By sliding the thumb across these points from right-to-left (Pinky to Index), a Tarbit maps exactly 16 values (0x0 to 0xF) on a single palm:

- **PINKY FINGER (0x0 - 0x3):** Base (0), Lower (1), Upper (2), Tip (3)
- **RING FINGER (0x4 - 0x7):** Base (4), Lower (5), Upper (6), Tip (7)
- **MIDDLE FINGER (0x8 - 0xB):** Base (8), Lower (9), Upper (A), Tip (B)
- **INDEX FINGER (0xC - 0xF):** Base (C), Lower (D), Upper (E), Tip (F)

This tactile system allows a scavenger to physically "hold" a hexadecimal memory address in their left hand in the dark, while their right hand operates a terminal or splices a live wire.

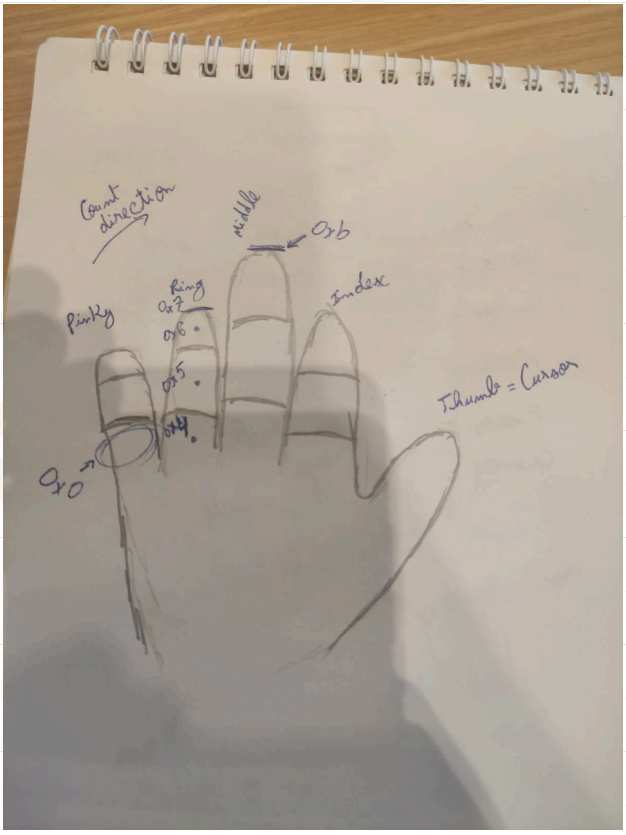


Figure 1: RECOVERED SCHEMATIC: THE BIOLOGICAL HEX-GRID.
NOTE THE STARTING SECTOR (0x0) AT THE PINKY BASE AND ENDIANNESS TRACE.

> 4. ASSEMBLY SYNTAX (V-O-O)

A Arta sentence operates like an Assembly Language instruction (MOV DEST, SRC). The syntax strictly follows a Verb-Object-Object structure. The Subject (the one executing the command) is not a separate word ; it is a one-letter prefix attached directly to the verb.

[THE SUBJECT PREFIXES]

- [m.] = I / Me (First Person)
- [n.] = We (First Person Plural)
- [k.] = You / Target (Second Person)
- [sh.] = He / She / It / They (External Pointer)

[SENTENCE ASSEMBLY]
To build a sentence, you attach the prefix to the root verb, followed by the target objects.

> [PREFIX].[VERB] [OBJECT_1] [OBJECT_2]

Example: If [t’ ə r] is “to build”, and [ŋ i] is “power”:

- [m.t’ ə r] [ŋ i] = “I build power.”
- [k.k’ ə s] [ŋ i] = “You kill the power.”

> 5. THE EXECUTION QUEUE (TIMELINE FLAGS)

In Arta, time is not a fluid, philosophical concept; it is a system log. A Tarbit does not speak in past, present, or future tenses. Instead, they treat time as an **Execution Queue**.

To indicate when an action happens, a Tarbit prepends the entire instruction with a **Timeline Flag**—a Power State bound to a Decimal Point (.)—acting as a header byte for the data packet. If no flag is present, the action defaults to the Present (Executing).

[THE QUEUE STATES]

- **[i.] (High State / Queued):** The instruction is scheduled. Represents the Future Tense.
- **[ə.] (Neutral State / Active):** The instruction is executing now. Represents the Present Tense.
- **[u.] (Low State / Logged):** The instruction has concluded and is cached. Represents the Past Tense.

[TIMELINE ASSEMBLY] The timeline flag is spoken as a sharp, standalone whisper before the main command.

> [FLAG] [PREFIX].[VERB] [OBJECT]

Examples:

- [i.] [m.t’ ə r] [a r t’ a] = “Target Queue: I will build the Arta system.”

- [u.] [k.k’ ə s] [ŋ i] = “Cached Log: You killed the power.”

> 6. LOGIC GATES (BRANCHING AND NEGATION)

A Tarbit cannot rely purely on linear execution; survival requires conditional logic. Arta uses “Dotted” Logic Gates to create IF/THEN statements and Nullifiers.

[THE COMPARE GATE (IF)]

To create a condition, append the [c.] (Relay Gate) modifier to the front of the phrase. The phrase is then followed by a Hardware Bus . to trigger the execution (THEN).

[THE INVERT GATE (NOT)]

To negate an object or an action, place the [x.] (Friction Gate) modifier immediately before the target word.

Example 1: Affirmative Condition > “If drone here then kill power.”

- [c.] [q' r i] . [m.k' a s] [ng i]
- (If: Drone [THEN] I-kill Power)

Example 2: Negative Condition

> “If no drone; no I kill power.”

- [c.] [x.] [q' r i] . [x.] [m.k' a s] [ng i]

- (Gate: Null Drone [THEN] Null I kill power)

[THE BOOLEAN GATES (AND / OR)]

To chain multiple objects or conditions together, Tarbits use the Boolean Dots. They sit strictly between the two elements they are modifying.

- [k'.] (The AND Gate): Binds two elements. Both must be true/present.
- [r.] (The OR Gate): Branches two elements. Either can be true/present.

Example 1: Boolean AND > “I scavenge the drone and the sensor.”

- [m.k' a x] [q' r i] [k'.] [c i t']
- (I scavenge . Drone [AND] Sensor)

Example 2: Complex Logic Branching > “If drone OR sensor here; kill power.”

- [c.] [q' r i] [r.] [c i t'] . [k' a s] [ng i]
- (Gate: Drone [OR] Sensor . kill power)

> 7. VECTOR OFFSETS (SPATIAL POINTERS)

Arta does not have prepositions; it uses **Vector Offsets.** When a Tarbit needs to specify a physical location, they place a Directional Dot immediately before the target object, treating the target as a base memory address and the dot as the spatial offset.

[THE OFFSETS]

- [f.] (Vent): Inside / Into
- [h.] (Exhaust): Outside / Out of
- [s.] (Static): Above / Over
- [ng.] (Drone): Below / Under
- [t'.] (Switch): At / Pinned to

Example 1: Internal Location > "I hide in the sanctuary."

- [m.q' u] [f.] [q' a]
- (I lock . [Vector In] Sanctuary)

Example 2: Grounded Location > "The coolant is under the drone."

- [f u x] [ng.] [q' r i]
- (Coolant . [Vector Low] Drone)

> 8. DATA POLLING (QUESTIONS)

A Tarbit does not use question words like "Who, What, Where, or Why." They execute a Data Poll. By placing the [q'.] (Query Flag) at the beginning of a phrase, the sentence becomes a GET request waiting for a return variable.

[POLLED EXECUTION] > "Did the drone kill the power?"

- [q'.] [u.] [sh.k' a s] [ng i]
- (Query . Past . It kills Power?)

[SPATIAL POLLING (WHERE)] To ask "Where," a Tarbit simply polls a Vector Offset. > "Where is the sanctuary?"

- [q'.] [f.] [q' a]
- (Query . Inside . Sanctuary?)

> 9. OPERATOR OVERLOADING (THE ALU MODIFIERS)

The Arithmetic Logic Unit (ALU) symbols do more than just math. When placed before a word, they act as parameter modifiers. Their exact physical translation depends entirely on the data type (Noun vs. Verb) they are modifying.

[THE DECREMENT (-)]

- **Applied to an Opcode (Throttle):** Executes the action slowly, quietly, or with low power.
  - - [m.k' a x] (I slowly scavenge)
- **Applied to a Register (Scale Down):** Modifies the object to be small, light, or minor.
  - - [s i r] (A small wire)

[THE INCREMENT (+)]

- **Applied to an Opcode (Overclock):** Executes the action rapidly, aggressively, or with high power.
  - + [m.f i r] (I run fast)
- **Applied to a Register (Scale Up):** Modifies the object to be large, heavy, or major.
  - + [q' r i] (A massive drone / A heavy drone)

[THE MULTIPLIER (*)]

- **Applied to an Opcode (Loop):** Executes the action repeatedly or continuously.
  - * [m.h i sh] (I repeatedly broadcast)
- **Applied to a Register (The Absolute Plural):** Allocates an array of multiple instances. This serves as the universal plural marker.
  - * [t' a r] (Builders)
  - * [q' r i] (A drone swarm / Many drones)

> 10. DATA ALLOCATION (NAMESPACES AND POINTERS)

Just as ALU operators change their function depending on what they modify, Arta overloads Subject Prefixes and Vector Offsets to manage physical inventory and spatial pointing.

[NAMESPACES (POSSESSION)]

To claim possession of a physical object, a Tarbit treats their Subject Prefix as a namespace. Attaching a Subject Prefix directly to a Noun (instead of a Verb) assigns ownership of that register.

- [m.] (My)
- [n.] (Our)
- [k.] (Your)
- [sh.] (Their / Its)

Example: Possession > “I scavenge your wire.”

- [m.k' a x] . [k.s i r]
- (I-scavenge [THEN] Your-wire)

[MEMORY POINTERS (DEMONSTRATIVES)]

When a Tarbit needs to distinguish between two identical objects in physical space (This vs. That), they overload the In/Out Vector Offsets to act as local and external memory pointers.

- [f.] (Local Pointer / THIS): The object is “In” the immediate vicinity or in hand.
- [h.] (External Pointer / THAT): The object is “Out” of reach or distant.

Example: Pointing > “This tool is broken. I need that tool.”

- [x.] [f.t' u c] . [m.c i r] [h.t' u c]
- (Null This-tool [THEN] I-fetch That-tool)

THE SCAVENGER'S SCRIPT (ORTHOGRAPHY)

In the Wastes, there is no paper and there is no ink. Tarbits write with screwdrivers, combat knives, and soldering irons on rusted server chassis and blast doors. Because carving curves into metal is slow, loud, and leaves jagged edges, the physical Arta script is entirely angular.

> 1. THE 7-SEGMENT SCRAPE

The alphabet is directly repurposed from the most common piece of intact Old World e-waste: the 7-segment digital LED display. Every character in Arta is formed by carving a specific combination of these 7 straight lines.

> 2. WHITESPACE AND THE EXECUTION TRIGGER (DP)

In the Wastes, carving takes energy. Therefore, standard spaces between words are simply left as empty, uncarved metal. The compiler ignores them. However, the standalone Decimal Point (DP) serves a highly specific, active hardware function:

1. **The Subject Prefix:** When carved within the same box as a Subject Glyph ([m], [n], [k]), it binds the subject to the verb.
2. **The Execution Trigger (THEN):** When the DP sits alone in its own character box ., it is not a space. It is the **Hardware Bus**. It acts as a logical THEN gate, signaling the end of a conditional statement ([c.]) and triggering the execution of the following command.

[THE PREFIX GLYPHS]

- **[m] (I/Me):** (Low bucket shape)
  - **As Prefix:**
- **[n] (We):** (High arch shape)
  - **As Prefix:**
- **[k] (You/Target):** (Left bracket shape)

- **As Prefix:**

> 3. THE GLYPH ARCHITECTURE

The shape of the carve reflects the acoustic nature of the sound.

[THE HARDWARE CONSONANTS]

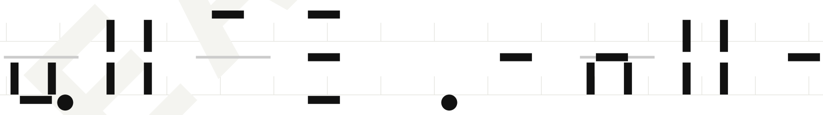
- **[t’] (Sharp Switch):** (Vertical dual-line structure).
- **[ʃ] (Static):** (Three parallel horizontal lines).
- **[q’] (Heavy Breaker):** (A heavy box structure).

[THE POWER STATES (VOWELS)] The power states map to the physical vertical height on the grid.

- **[i] (High State):** (Top bar / High voltage).
- **[ə] (Neutral State):** (Middle bar / Baseline).
- **[u] (Low State):** (Bottom bar / Grounded).

> 4. FIELD EXECUTION: “I BOOT THE SYSTEM”

To execute the command [m.t’ i ʃ] [ə r t’ ə] on a terminal or blast door, a Tarbit utilizes the DP-Space protocol to ensure the verb and object are distinct while keeping the subject-prefix attached.



To anyone else, it looks like a meaningless series of scratches and a bullet hole. To a Swapper, it is a perfectly formatted initialization command for the core Arta logic.

APPENDIX A: ARTA SYSTEM CHARACTER MAP

This appendix serves as the master register for field-scraping and terminal-emulation. It provides a 1:1 mapping between acoustic hardware signals and their physical 7-segment scrape patterns.

> 1. THE ACOUSTIC OPCODES (CONSONANTS)

[t’]

[k’]

[q’]

[ǀ]

[S]

[ʃ]

[x]

[f] / [h]

[ʁ]

[r]

[ŋ]

> 2. THE VOLTAGE FLAGS (VOWELS)

[i] (High)

[ə] (Neutral)

[u] (Low)

> 3. THE SUBJECT REGISTERS (PREFIXES)

[m] + DP

[n] + DP

[k] + DP

> 4. TIMELINE FLAGS (EXECUTION QUEUE)

[i.] (Queued)

[ə.] (Active)

[u.] (Logged)

> 5. LOGIC GATES (BRANCHING AND NEGATION)

[c.] (IF)

[x.] (NOT)

> 5. VECTOR OFFSETS

[f.] (IN)

[h.] (OUT)

[s.] (Above)

[ng.] (Below)

[t’] (at)

> 5. THE ALU REGISTERS (MATH OPERATORS)

Used for base-16 calculation, memory address incrementing, and variable assignment.

+ (Add)

- (Sub)

***** (Mult)

/ (Div)

= (Assign)

> 6. HEXADECIMAL NUMBERS (0X0 - 0XF)

Numbers follow standard 7-segment numeric mapping to ensure compatibility with Old World hardware displays.

0x0

0x1

0x2

0x3

0x4

0x5

0x6

0x7

0x8	**0x9**	**0xA**	**0xB**
8	9	A	b
0xC	**0xD**	**0xE**	**0xF**
C	d	E	F

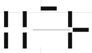
THE SCAVENGER'S LEXICON: COMPILED ARCHIVE █

THE LEXICON MEMORY MAP
The Arta dictionary is not sorted alphabetically; it is sorted by hardware execution priority. The logic system loads baseline acoustic signals first (Opcodes and Power States). Once the physical hardware is established, execution crosses the **DP Boundary** to process syntax modifiers (The “Dotted” Registers and Flags).

SECTOR	HARDWARE CLASS	RAW INPUTS
0x0	**Grid Hums**	m, n, k
0x1	**Relays & Switches**	t', k', q', c
0x2	**Static & Vents**	s, sh, x, f, h, gh
0x3	**Failing Motors**	r, ng
0x4	**Power States**	i, a, u
	— THE DP BOUNDARY (MODIFIERS) —	
0x5	**Subject Registers**	m., n., k.
0x6	**Timeline Flags**	i., a., u.
0x7	**Logic Gates**	c., x.
0x8	**Vector Offsets**	t'., s., f., h., ng.
0x9	**The Hardware Bus**	.

> SECTOR [T']

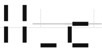
t' i k' *t'ik'*



noun: A logical pointer directing execution to a specific hardware address.

---

t' u c *t'ul*



noun: A physical, hand-held maintenance tool.

> SECTOR [K’]

k' i x *k'ix*

verb: To permanently flash or burn data onto a hardened ROM chip.

---

k' a x *k'əx*

verb: To physically strip or scavenge a hardware component from a larger system.

> SECTOR [Q’]

q' r i *q'ri*

noun: An active, autonomous Martas defense drone.

---

q' a *q'ə*

noun: A physically insulated, offline sanctuary safe from network intrusion.

---

q' u t' *q'ut'*

verb: To logically partition or reserve a sector of raw storage.

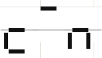
> SECTOR [C]

cit' *ǀit'*

noun: An environmental or industrial hardware sensor (e.g., accelerometer or gyro).

---

cir *ǀir*



verb: To fetch or read raw data from a specific memory address.

---

cac ǀəǀ

verb: To physically splice or bridge two live electrical wires.

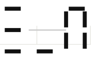
> SECTOR [S]

sir *sir*

noun: A physical copper wire or data bus.

---

sung *suŋ*



noun: A temporary data buffer or holding queue.

> SECTOR [X]

xish *xiʃ*

noun: A mid-frequency radio transmission or broadcast signal.

---

xa *xə*

noun: An empty, unwritten sector of memory; a null pointer.

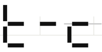
> SECTOR [F]

fir *fir*

verb: To execute a continuous software process or loop.

---

fac *fəl*



verb: To securely flush a buffer or drop a software connection.

---

fux *fux*

noun: Liquid coolant or industrial fluid used in thermal management.

> SECTOR [H]

hish *hiʃ*

verb: To echo or output a raw signal through a hardware terminal.

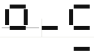
---

hush *huʃ*

verb: To place a system into a low-voltage idle or standby state.

> SECTOR [GH]

gh a sh *ʁəʃ*



verb: To aggressively overclock a hardware component past its stable baseline.

> SECTOR [A]

a r t' a *ərt'ə*

noun: The root execution language and underlying logic of the enclave.

> SECTOR [M.]

m. *m.*

prefix: [FIRST PERSON] Binds the opcode execution to the speaker (I/Me).

> SECTOR [N.]

n. *n.*

prefix: [FIRST PLURAL] Binds the opcode execution to the speaker's group (We).

> SECTOR [K.]

k. *k.*

prefix: [SECOND PERSON] Binds the opcode execution to the target entity (You).

> SECTOR [SH.]

sh. *ʃ.*

prefix: [EXTERNAL POINTER] Binds the opcode execution to a third-party or unidentified entity (He/She/It/They).

> SECTOR [C.]

c. l.

modifier: [GATE] The Relay dot. Evaluates the following phrase as a condition (IF).

> SECTOR [X.]

x. *x.*

modifier: [NOT] The Friction dot. Inverts or nullifies the following opcode or register (NO/FALSE).

> SECTOR [Q’.]

q'. *q'.*

modifier: [QUERY FLAG] The Breaker dot. Casts the following phrase as a polling request for data (Question).

> SECTOR [T’.]

t'. *t'.*

modifier: [VECTOR PIN] The Switch dot. Indicates an exact, physical connection or adjacent location (at/on).

> SECTOR [S.]

s. *s.*

modifier: [VECTOR HIGH] The Static dot. Indicates location above, on top of, or at a higher physical elevation.

> SECTOR [F.]

f. *f.*

modifier: [VECTOR IN] The Vent dot. Indicates movement into or location inside a structure or port.

> SECTOR [H.]

h. *h.*

modifier: [VECTOR OUT] The Exhaust dot. Indicates movement out of or location external to a structure.

> SECTOR [NG.]

ng. *ŋ.*

modifier: [VECTOR LOW] The Drone dot. Indicates location below, underneath, or grounded beneath an object.

> SECTOR [I.]

i. *i.*

modifier: [QUEUED] The High-State dot. Indicates an action scheduled to happen (Future).

> SECTOR [A.]

a. *ə.*

modifier: [ACTIVE] The Neutral-State dot. Indicates an action currently executing (Present).

> SECTOR [U.]

u. *u.*

modifier: [LOGGED] The Low-State dot. Indicates an action that has concluded and written to memory (Past).
